

OELLM

Observation of Earth Large Language Model

A Vision–Language Model for Urban Understanding &
Cross-View Geolocalisation

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The problem: VLMs can't really “read” a place

Vision–Language Models are tested on web images, not on geography. Two abilities stay unmeasured:

- **Urban understanding** — land use, building stock, transit, green space from one location.
- **Cross-view geolocalisation** — do this *satellite* view and this *street* view show the same place? From which direction?

The trap

Unimodal shortcut learning: the model answers from a text prior or one image — never fusing the two views.



Samsun, Türkiye - topic: camera_direction - difficulty: medium - answer: B

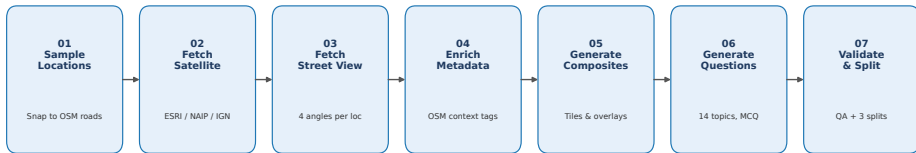
In which satellite image does the red arrow point in the direction this street view was captured?

- A. Image A
- B. Image B [correct]**
- C. Image C
- D. Image D

One satellite tile + four street views per location.

Our answer: a zero-hallucination, OSM-grounded dataset

EOLLM pipeline — seven generation stages



- Every answer derived **deterministically from OpenStreetMap** — no annotators, bit-exact reproducible.
- **40 cities**, 32 countries, 6 continents.
- **39,155** question records over **4,168** locations.
- **14 task types** → urban understanding + cross-view geometry.

Method: a compact model, parameter-efficient tuning

Model. Fine-tune **Qwen3.5-4B-VL** with **LoRA** (vision + language, all-linear), bf16, on a single RTX PRO 6000.

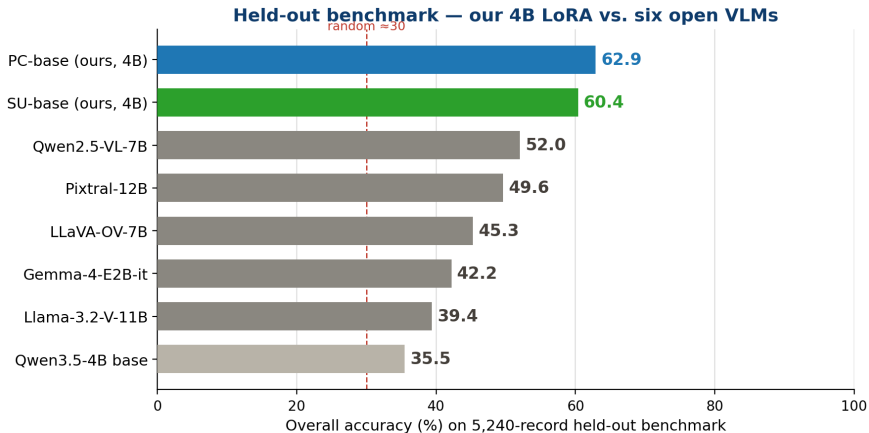
Eval. Strict **greedy decoding**, deterministic, scored against per-topic *random* and *majority* baselines — not just chance.

Three splits, one taxonomy:

Split	Train	Tests
per_city	26,931	new locations
seen_unseen	26,963	8 held-out cities
benchmark	—	disjoint, held-out

seen_unseen is the honest test of geographic generalisation.

Result: a 4B model beats six open VLMs



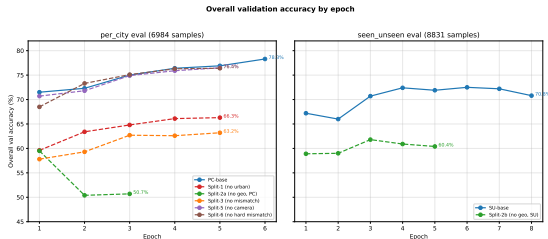
5,240-record held-out benchmark · identical greedy protocol. Our **PC-base** and **SU-base** (4B) lead models up to **3× larger**.

The real finding: the skill transfers across cities

- **PC-base** (seen cities): **62.9%**
- **SU-base** (8 cities fully held out): **60.4%**

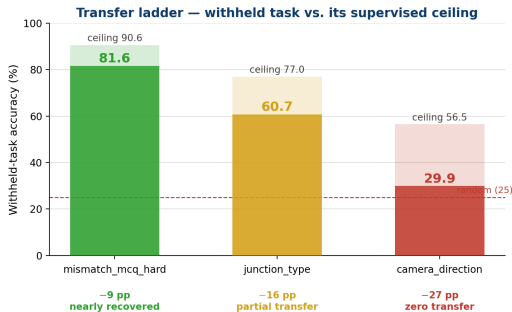
Only a 2.5-point gap

Holding out *entire cities* costs almost nothing — the model learns **transferable urban structure**, not memorised places.



Validation accuracy by epoch: per-city (left) vs. cross-city (right).

Ablations: which abilities transfer, which don't



We withhold whole task families from training. Three clean regimes:

- **Within-family** — never-seen hard mismatch still hits 81.6% (easy carries hard).
- **Cross-family** — urban tasks recover $\sim 2/3$ of ceiling from geometry alone.
- **Zero transfer** — camera direction stays at chance unless trained directly.

A map of what a VLM learns by analogy vs. needs taught explicitly.

Impact

- A **4B model on one GPU** that beats 12B general VLMs — cheap, scalable urban analytics.
- Open release: **model adapters, dataset, benchmark** + evaluation harness.
- A reproducible reference point for geospatial VLMs.

Next steps

- Second-generation dataset with tighter distributional balance.
- **Reinforcement learning** with verifiable rewards to lift the tasks that resist transfer.
- Scale beyond 40 cities; richer street-level inputs.

OELLM

A 4B vision–language model that reads cities —
and **generalises** to ones it has never seen.

PC-base **62.9%** · SU-base **60.4%** · beats 6 open VLMs

Thank you — questions?